

Freight Rate Prediction — ML Assessment Executive Summary Predict posted_rate (USD) for freight loads. Final model: HistGradientBoosting with feature set Q and log1p target. Trained on 48,000 Jan–Oct rows; inferred 12,000 Nov–Dec holdout rows. Development chronological validation MAE: 106.83 (primary) and 113.29 (sensitivity). These are NOT Nov–Dec holdout scores. score.py validates format only (exit code 0).

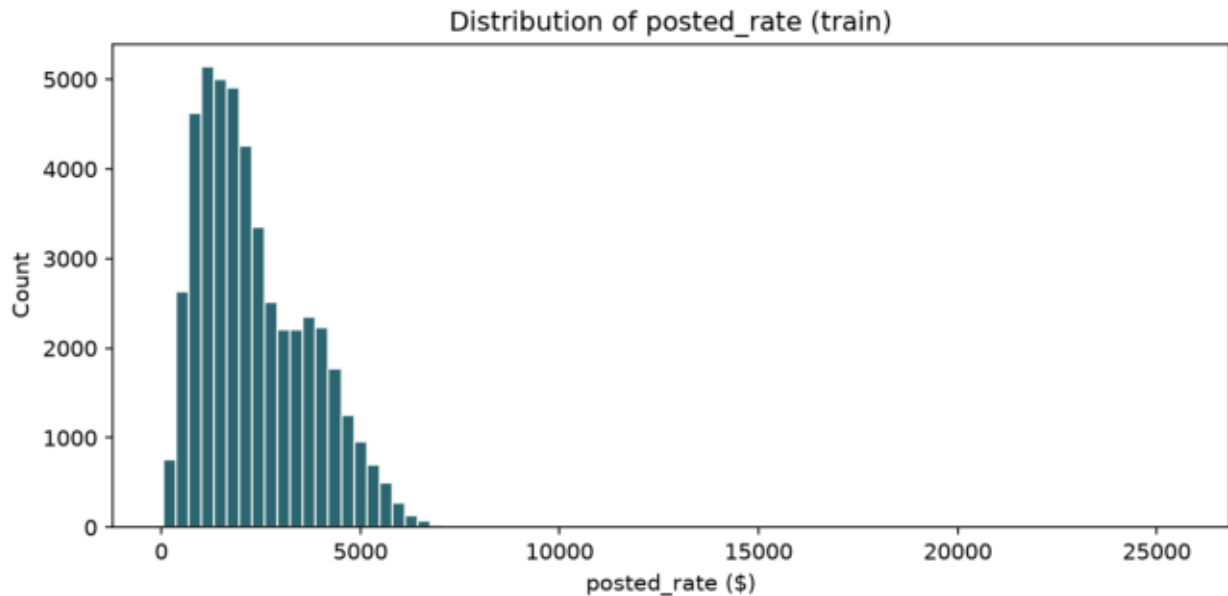
Dataset & Validation Strategy train-test.csv: 48,000 labeled rows. validation.csv: 12,000 unlabeled holdout. Primary split: Jan-Aug train → Sep-Oct validate. Sensitivity: Jan-Sep train → Oct validate. No random shuffle. Preprocessing fit on training fold only.

Feature Set Q & Model Selection Features: distance, log_distance, distance_bin, weight, weight_is_missing, quote_signal, equipment. Excluded: raw market_index, route/city OHE, geographic extras. Hyperparameters: max_depth=6, l2=0.1, lr=0.08, max_iter=300, random_state=42.

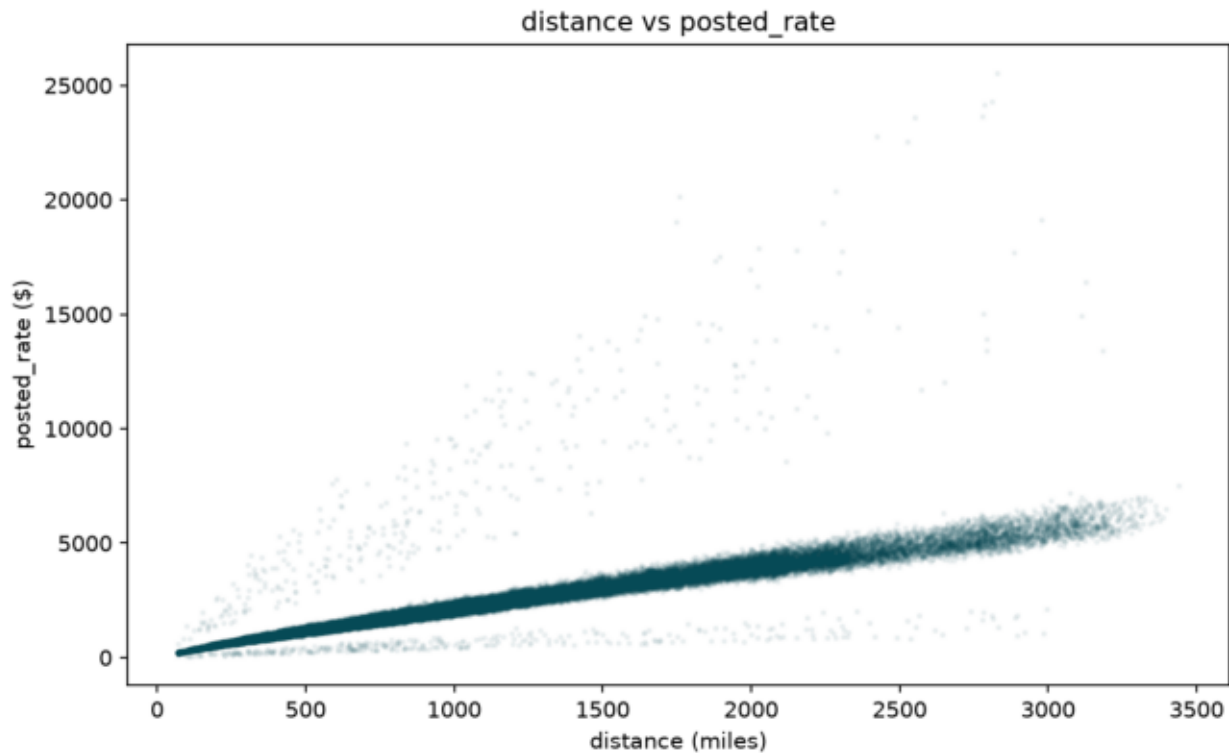
Model Comparison (development primary MAE) Global median: 1148.92 | Distance linear: 196.95 | Ridge: 148.06 HistGB FULL: 129.69 | HistGB log1p FULL: 118.83 HistGB Q log1p (final): 106.83 primary / 113.29 sensitivity Final predictions: mean \$2345.93, median \$2026.45, min \$210.94, max \$6548.41.

Limitations • No local Nov-Dec ground truth • score.py does not compute accuracy • Long-tail underprediction remains • CatBoost not evaluated • December chart flat at \$841.48 (Q excludes calendar features)

posted_rate distribution

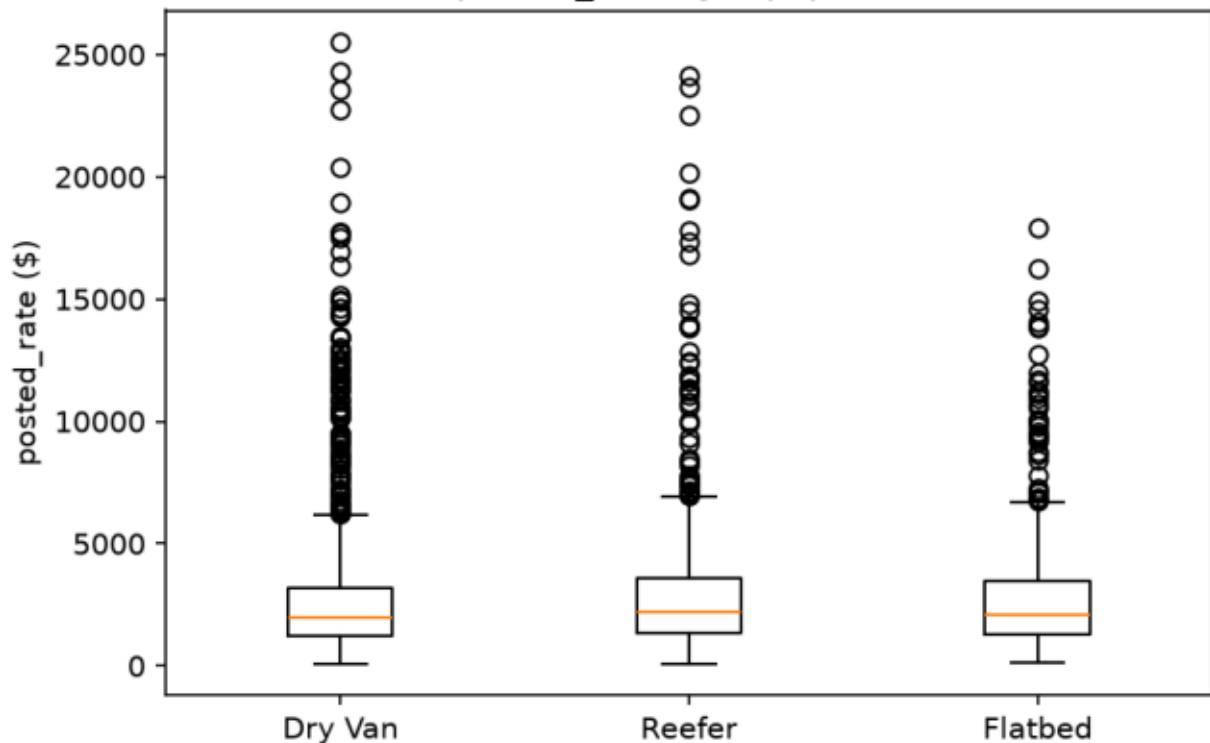


Distance vs posted_rate



Rate by equipment

posted_rate by Equipment



December 2025 candidate prediction

Candidate: December 2025 Predicted Load Rate

